

Unit 8, Lesson 7: Volume of Rectangular Prisms

Benchmark

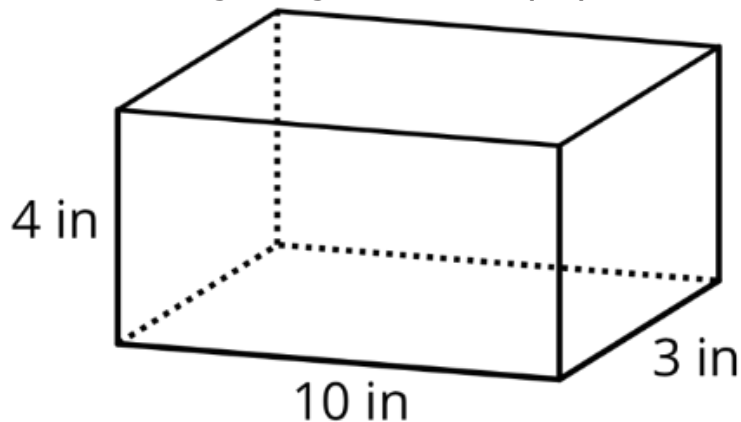
MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.

Learning Target

- ☐ I can find the volume of a rectangular prism.

8.7.1 Warm-Up: A Box of Cubes

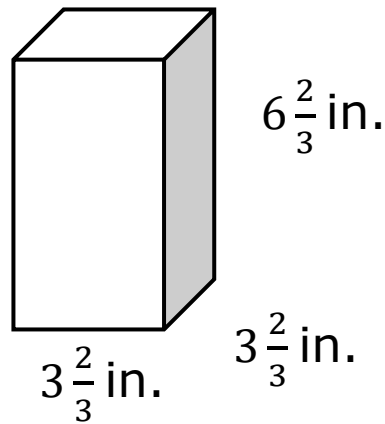
1. How many cubes with an edge length of 1 inch (in.) fill the box?



2. If the cubes had an edge length of 2 inches, explain if more or fewer cubes would be needed to fill the box.
3. If the cubes had an edge length of $\frac{1}{2}$ inch, explain if more or fewer cubes would be needed to fill the box.

8.7.2 Guided Instruction: Two Methods for Finding the Volume of a Right Rectangular Prism

Alan calculated the volume of the rectangular prism.



1. Fill in the blanks to complete Alan's statement on how he calculated the volume.

To find the volume of this prism, I will fill the prism with cubes of side lengths $\frac{1}{3}$ in.

Because $3\frac{2}{3} \div \frac{1}{3} = \underline{\hspace{2cm}}$, $\underline{\hspace{2cm}}$ cubes will fit along the length and $\underline{\hspace{2cm}}$ cubes will fit along the width. $\underline{\hspace{2cm}}$ cubes will cover the base of the prism. The height of the prism can be used to determine how many layers of cubes are needed. It will take $\underline{\hspace{2cm}}$ layers of cubes because $6\frac{2}{3} \div \frac{1}{3} = \underline{\hspace{2cm}}$. In total, it will take $\underline{\hspace{2cm}}$ of the $\frac{1}{3}$ in. cubes to fill the prism.

The volume of each small cube is $\frac{1}{27}$ in³.

2. Multiply the number of cubes needed to fill the prism by the volume of each cube to find the volume of the prism.

3. Jai used the formula $V = l \times w \times h$ to find the volume of the prism. Complete Jai's work to find the volume of the prism.

$$V = l \times w \times h$$

$$V = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$$

$$V = \underline{\hspace{1cm}} \text{ in}^3$$

4. How are Alan and Jai's methods similar?



The volume, V , of a rectangular prism is

$V = l \times w \times h$, where l represents the length, w represents the width, and h represents the height. This can also be written as $V = B \times h$, where B represents the area of the base and h represents the height.

5. Discuss with your partner how the formula $V = B \cdot h$ is related to the formula $V = l \cdot w \cdot h$.

6. Complete the statements based on your discussion.

The product of the length, l , and the width, w , is equal to

_____ of the prism.

$$B = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$$

Both formulas can be used to find the volume of a right rectangular prism.

8.7.3 Guided Practice: Volume of a Right Rectangular Prism

1. Diego correctly points out that 81 cubes with an edge length of $\frac{1}{3}$ centimeters (cm) are needed to fill a right rectangular prism that has length of 3 cm, width of 1 cm, and height of 1 cm.
 - a. Explain or show how this is true.
 - b. What is the volume, in cubic centimeters, of the rectangular prism?
2. Lin and Noah are packing small cubes into a cube with an edge length of $1\frac{1}{2}$ inches. Lin is using cubes with an edge length of $\frac{1}{2}$ inch, and Noah is using cubes with an edge length of $\frac{1}{4}$ inch.
 - a. Explain who would need more cubes to fill the $1\frac{1}{2}$ in. cube.
 - b. Explain whether Lin and Noah will find the same volume of the $1\frac{1}{2}$ inch cube using their method.
 - c. What is the volume of the cube with $1\frac{1}{2}$ inch edge lengths? Include units.

3. A toy company is packaging its toys to be shipped. Some small toys are placed inside a cube-shaped box with side lengths of 0.5 inches (in.). These smaller boxes are then packed into a shipping box with dimensions of 12 in. $\times$ 35 in. $\times$ 2.5 in.
- a. How many small toys can be packed into the larger box for shipping?

 - b. Use the number of toys that can be shipped in the box to determine the volume of the box. Include units.

8.7.4 Your Turn!

1. To give their animals essential minerals and nutrients, farmers and ranchers often have a block of salt, called a "salt lick" available for their animals. A rancher is ordering a box of cube-shaped salt licks. The edge lengths of each salt lick are $\frac{5}{12}$ foot (ft).



- a. Explain if the volume of one salt lick is greater than or less than 1 cubic foot (ft^3).
- b. A box that is $1\frac{1}{4}$ ft $\times$ $1\frac{2}{3}$ ft $\times$ $1\frac{5}{6}$ ft is used to ship the salt licks. How many cubes of salt lick can fit in the box?
2. A pool in the shape of a rectangular prism is being filled with water. The dimensions of the pool are 24 feet (ft) by 15 ft. If the height of the water in the pool is 1.3 ft, what is the volume of the water in cubic feet?

8.7.5 Wrap-Up: Ticket Out the Door

1. Joy keeps her modeling clay in a plastic box that has a length of 21 centimeters (cm), a width of 21 cm, and a height of 63 cm.
 - a. What is the volume of the plastic box? Include units.

 - b. If each piece of cube-shaped clay has a side length of 4.2 cm, how many pieces of clay can be stored in the box?